

Debasish Chakroborti (Joy), Ph.D.

Computer Science / Software Engineering
410 Stensrud Rd #118, Saskatoon, SK
<https://joydeba.github.io/> · debasish.chakroborti@usask.ca

August 2026

Dear Members of the Search Committee,

I am writing to apply for the Assistant Professor position in Computer Science / Software Engineering. I specialize in AI-assisted software maintenance, with a Ph.D. in Computer Science from the University of Saskatchewan (2024), ongoing postdoctoral research, industry collaborations, and substantial teaching experience across university and polytechnic settings. I look forward to contributing to a strong academic unit in software engineering, AI/ML for software systems, and related computing areas while building an externally funded research program and mentoring students.

My research aims to deliver issue-free software on the versions people actually use. Under Professors Chanchal K. Roy and Kevin Schneider, I developed Baczer, ReBack, BackSlice, and BackTrans—tools that combine repository mining, machine learning (including CNNs and LLMs), and program analysis to automate software backporting at scale. This work has appeared in venues including ICSE, ICPC, MSR, CASCON, and ASE Journal. Complementary contributions include scientific workflow data reusability (RISP/RISPs) and applied AI automation through my Living Skies Postdoctoral Fellowship (NSERC CREATE SOAR / related Alliance work) and consulting with Cloud Linux Software, Inc. Looking ahead, I plan to advance intelligent release engineering and quality assurance, AI-assisted developer workflows and technical debt, and industry-connected applied software engineering.

I bring a strong record of teaching and curriculum development. I currently teach in the School of Computing and Digital Innovation at Saskatchewan Polytechnic (Computer Systems Technology; Cloud Computing & Blockchain), and I recently served as Sessional Lecturer for CMPT 145 at the University of Saskatchewan with full course responsibility. Earlier, as Lecturer in CSE at Khulna University, I taught programming and software engineering and completed Foundation Training for University Teachers. My teaching emphasizes experiential learning, inclusive pedagogy, and contemporary practice. I have mentored an M.Sc. student (with co-authored publications), supervised undergraduate summer researchers, and guided teaching assistants. I am prepared to teach core courses in programming, software engineering, and systems, as well as topics such as software maintenance, mining software repositories, and AI for software engineering.

Beyond research and teaching, I am committed to service and community leadership, including as President of the Computer Science Graduate Council at the University of Saskatchewan, and I have completed Reconciliation Education training. I value collaborative academic cultures and look forward to contributing to departmental service, student success, and equity-minded practices.

Enclosed please find my curriculum vitae, research statement, and teaching statement. I would welcome the opportunity to discuss how my experience can support your department's mission. Thank you for your consideration.

Sincerely,

Debasish Chakroborti (Joy), Ph.D.